US012404150B2

(12) **United States Patent**
Chen et al.

(10) **Patent No.: US 12,404,150 B2**
(45) **Date of Patent: Sep. 2, 2025**

(54) **FULL-TIME ANTI-SWAY CONTROL METHOD OF BRIDGE CRANE SYSTEM BASED ON INVERTER STRUCTURE**

(71) Applicant: **Delta Electronics, Inc.**, Taoyuan (TW)

(72) Inventors: **Wan-Ping Chen**, Taoyuan (TW); **Chun-Ju Wu**, Taoyuan (TW)

(73) Assignee: **DELTA ELECTRONICS, INC.**, Taoyuan (TW)

(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 395 days.

(21) Appl. No.: **17/491,982**

(22) Filed: **Oct. 1, 2021**

(65) **Prior Publication Data**

US 2022/0227604 A1 Jul. 21, 2022

Related U.S. Application Data

(60) Provisional application No. 63/138,640, filed on Jan. 18, 2021.

(30) **Foreign Application Priority Data**

Sep. 14, 2021 (CN) 202111074834.3

(51) **Int. Cl.**

G06F 7/70	(2006.01)
B66C 13/06	(2006.01)
B66C 13/46	(2006.01)
G05B 13/04	(2006.01)

(52) **U.S. Cl.**

CPC ***B66C 13/063*** (2013.01); ***B66C 13/46*** (2013.01); ***G05B 13/042*** (2013.01); *B66C 2700/084* (2013.01); *G05B 2219/41217* (2013.01)

(58) **Field of Classification Search**

CPC . B66C 13/063; B66C 13/46; B66C 2700/084; G05B 13/042; G05B 2219/41217

See application file for complete search history.

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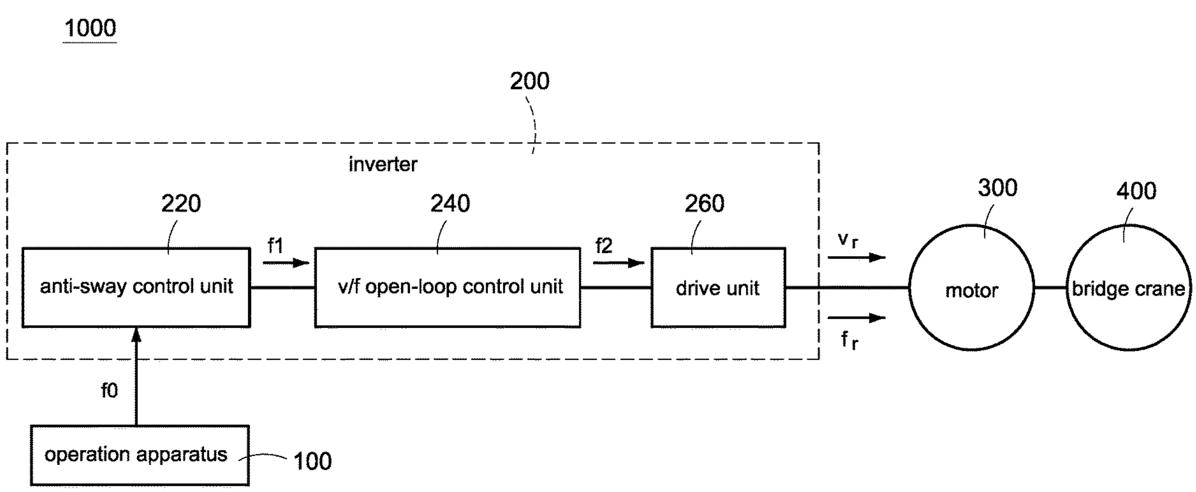
Primary Examiner — Adam M Alharbi

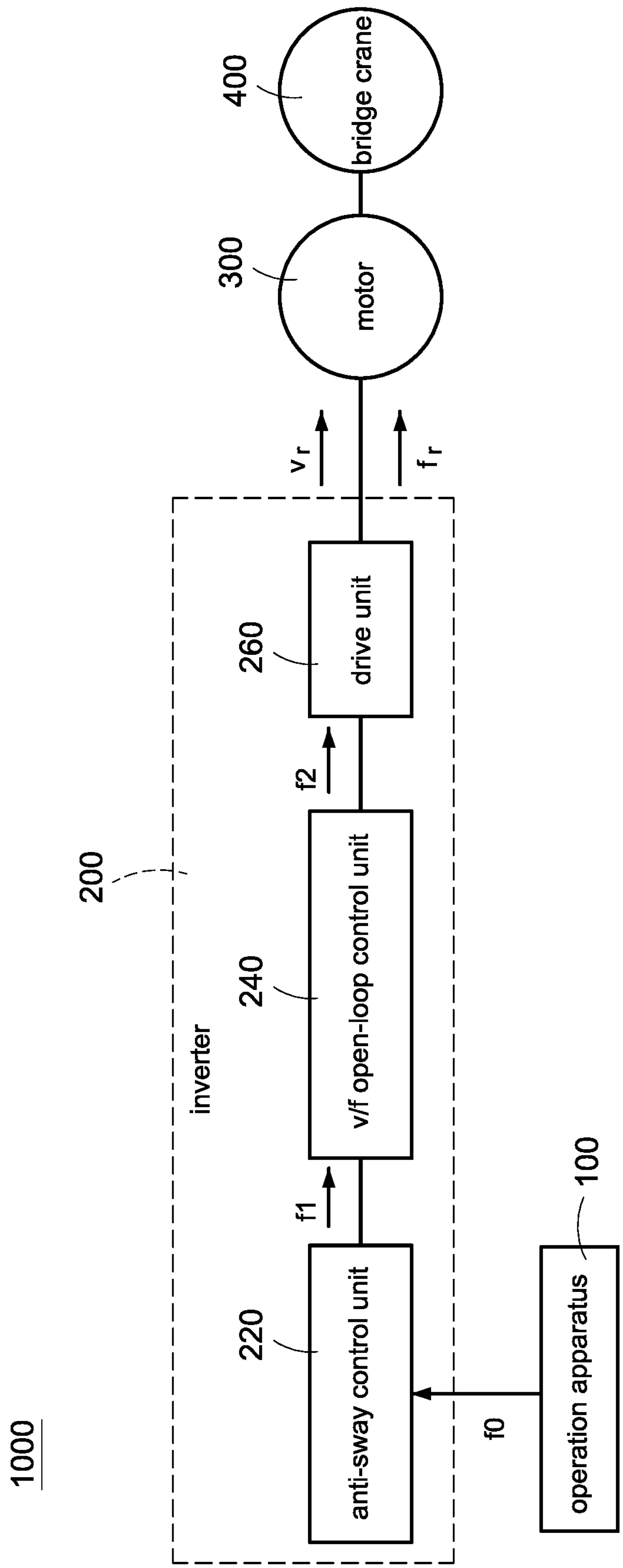
(74) *Attorney, Agent, or Firm* — Chun-Ming Shih; HDLS IPR SERVICES

(57) **ABSTRACT**

A full-time anti-sway control method of a bridge crane system based on an inverter structure includes steps of: receiving a specified high frequency and a frequency change time, calculating a time setting range according to a plurality of system parameters and a rope length information of the bridge crane system, selecting a time setting value within the time setting range, dividing the frequency change time into a plurality of time intervals according to the time setting value, adjusting an operation frequency command to change between a low frequency and the specified high frequency within the plurality of time intervals to generate a frequency change curve, calculating a frequency correction amount according to the frequency change curve and the rope length information, and superimposing the frequency change curve and the frequency correction amount to generate an anti-sway frequency command to drive the at least one motor.

14 Claims, 11 Drawing Sheets

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200
inverter
220
anti-sway control unit
f1
240
v/f open-loop control unit
f2
260
drive unit
v_r
f_r
300
motor
400
bridge crane
f0
operation apparatus
100

FIG.1

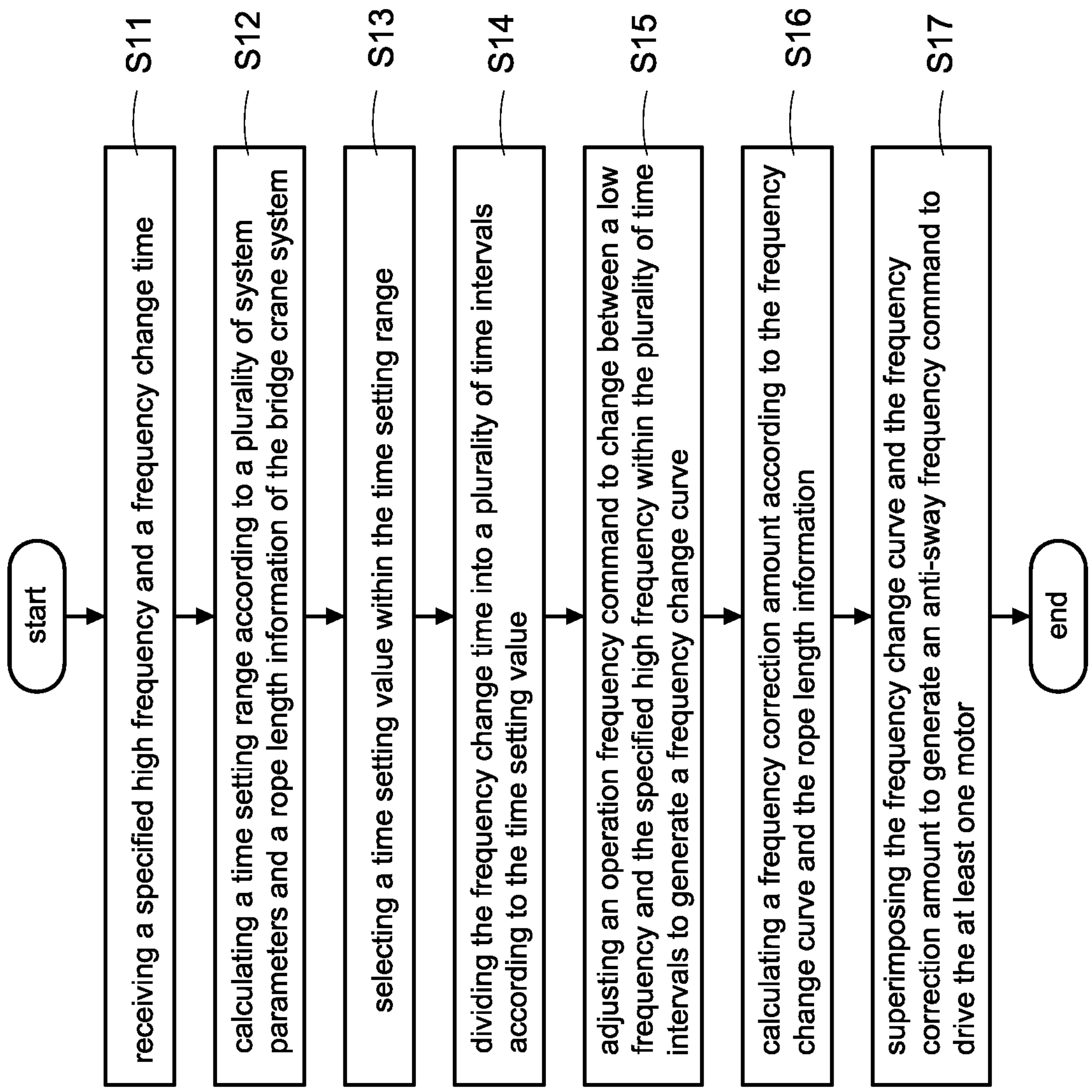

start
receiving a specified high frequency and a frequency change time
S11
calculating a time setting range according to a plurality of system parameters and a rope length information of the bridge crane system
S12
selecting a time setting value within the time setting range
S13
dividing the frequency change time into a plurality of time intervals according to the time setting value
S14
adjusting an operation frequency command to change between a low frequency and the specified high frequency within the plurality of time intervals to generate a frequency change curve
S15
calculating a frequency correction amount according to the frequency change curve and the rope length information
S16
superimposing the frequency change curve and the frequency correction amount to generate an anti-sway frequency command to drive the at least one motor
S17
end

FIG.2

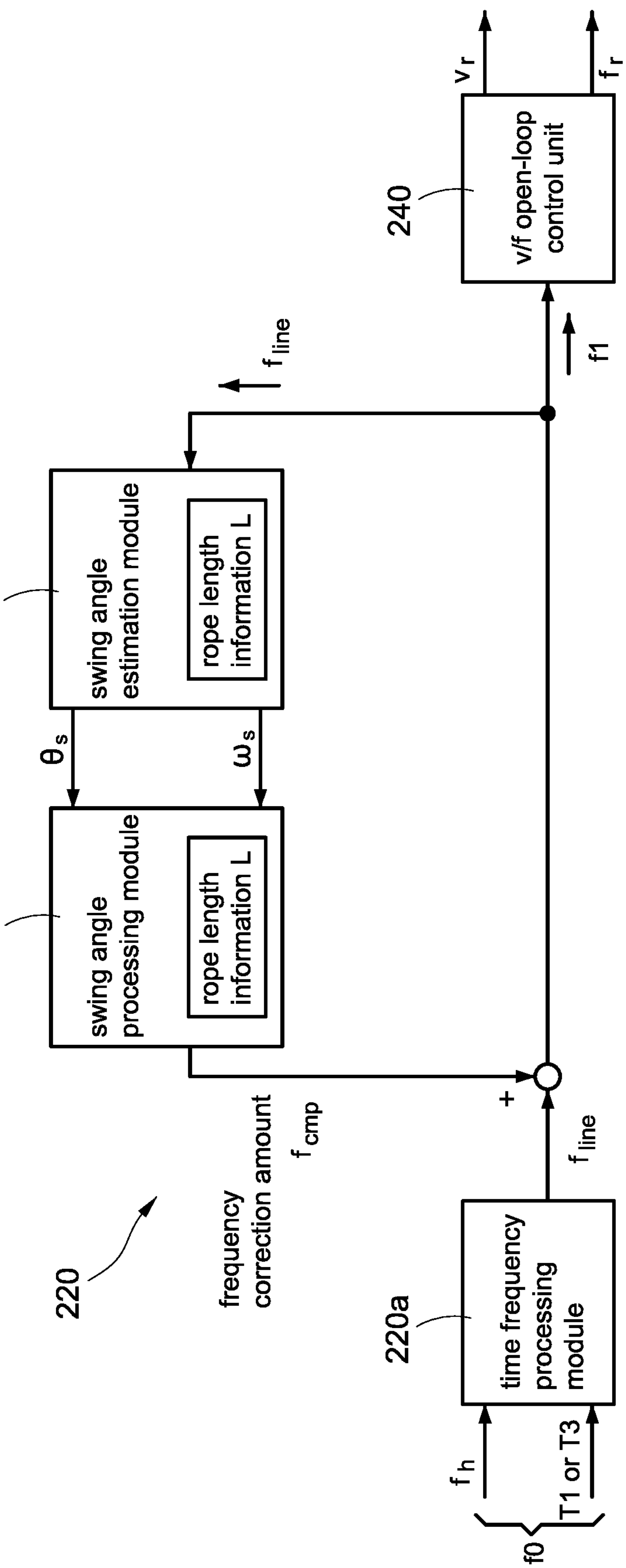

240
v/f open-loop control unit
v_r
f_r
f1
f_{line}
220b
swing angle estimation module
rope length information L
θ_s
ω_s
220c
swing angle processing module
rope length information L
frequency correction amount f_{cmp}
+
+
f_{line}
220
220a
time frequency processing module
f_h
T1 or T3
f0

FIG.3

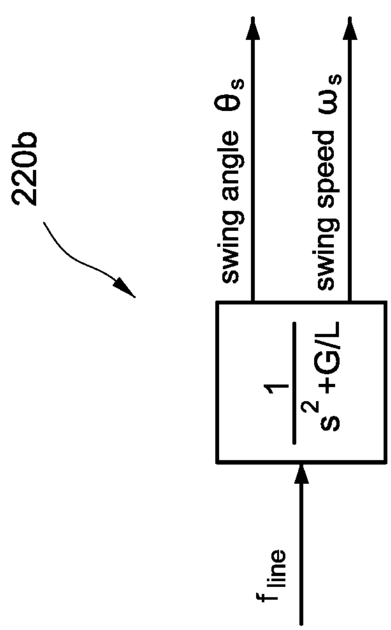

220b
f_{line}
$\frac{1}{s^2+G/L}$
swing angle θ_s
swing speed ω_s

FIG.4A

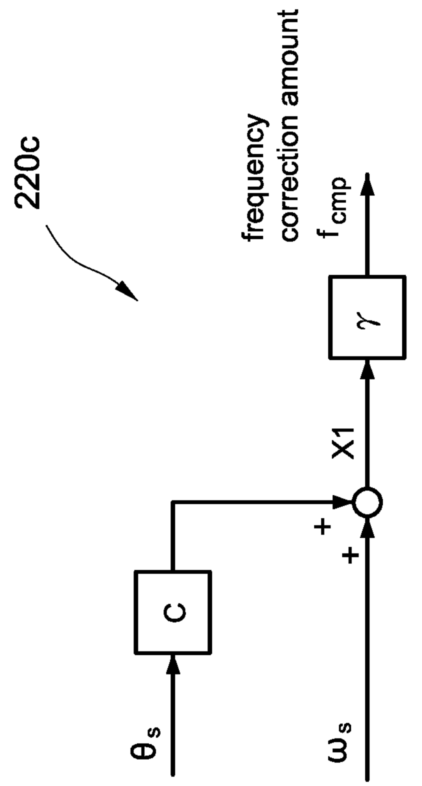
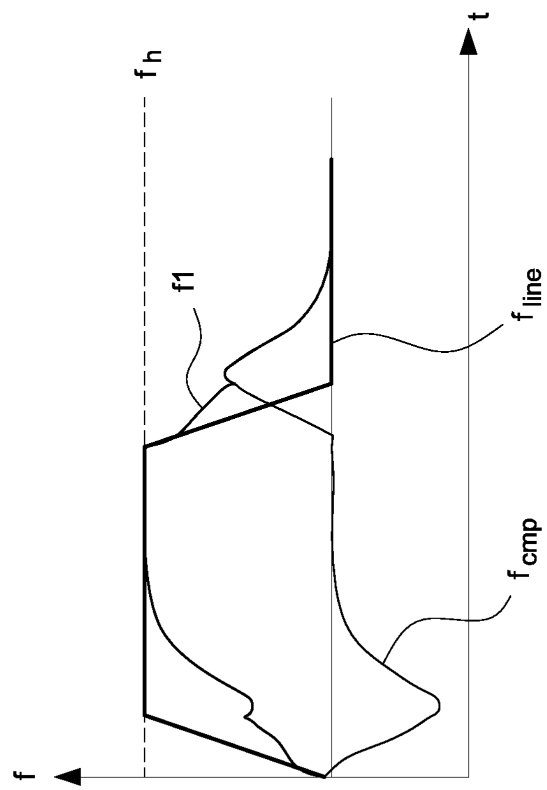
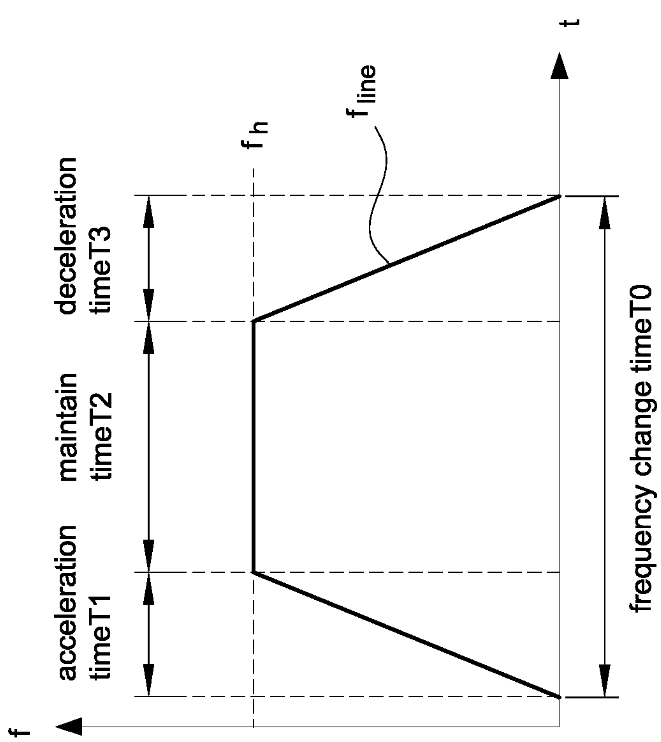

220c
θ_s
C
ω_s
+
+
X1
γ
frequency
correction amount
f_{cmp}

FIG.4B



f
t
f_h
f1
f_{line}
f_{cmp}

FIG.5B



f
t
f_h
f_{line}
acceleration timeT1
maintain timeT2
deceleration timeT3
frequency change timeT0

FIG.5A

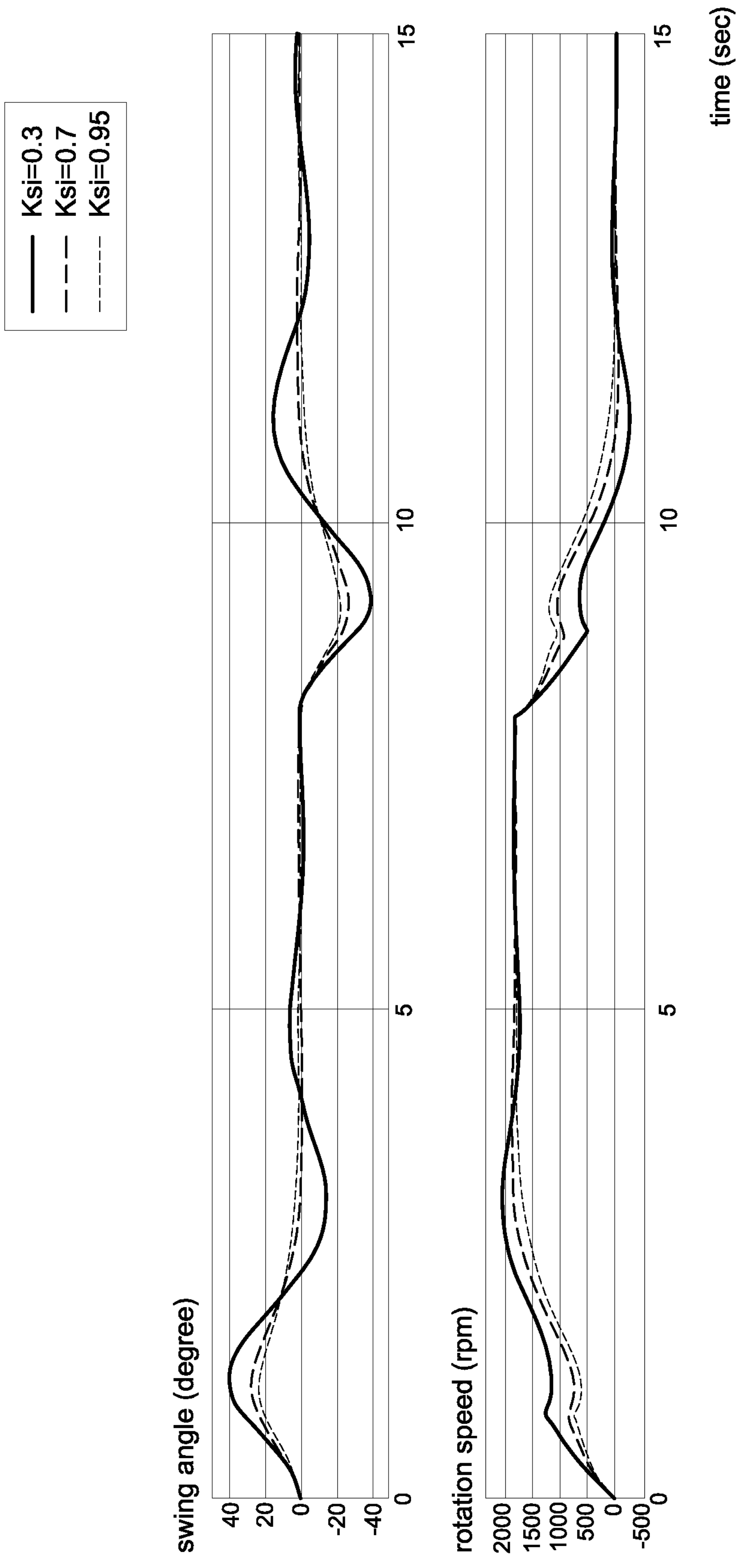

Ksi=0.3
Ksi=0.7
Ksi=0.95
swing angle (degree)
40
20
0
-20
-40
rotation speed (rpm)
2000
1500
1000
500
0
-500
0
5
10
15
time (sec)

FIG.6

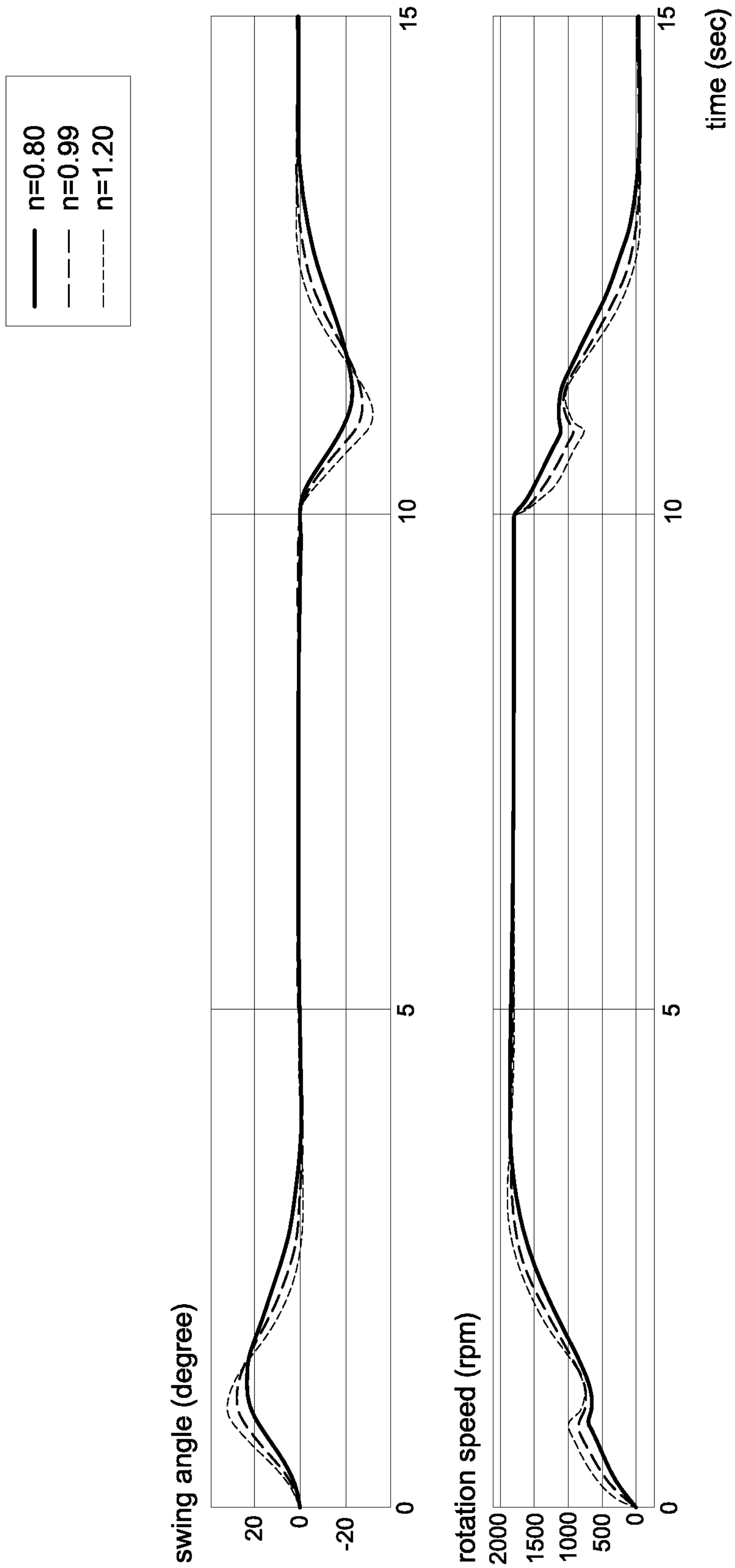

n=0.80
n=0.99
n=1.20
swing angle (degree)
20
0
-20
rotation speed (rpm)
2000
1500
1000
500
0
0
5
10
15
time (sec)

FIG.7

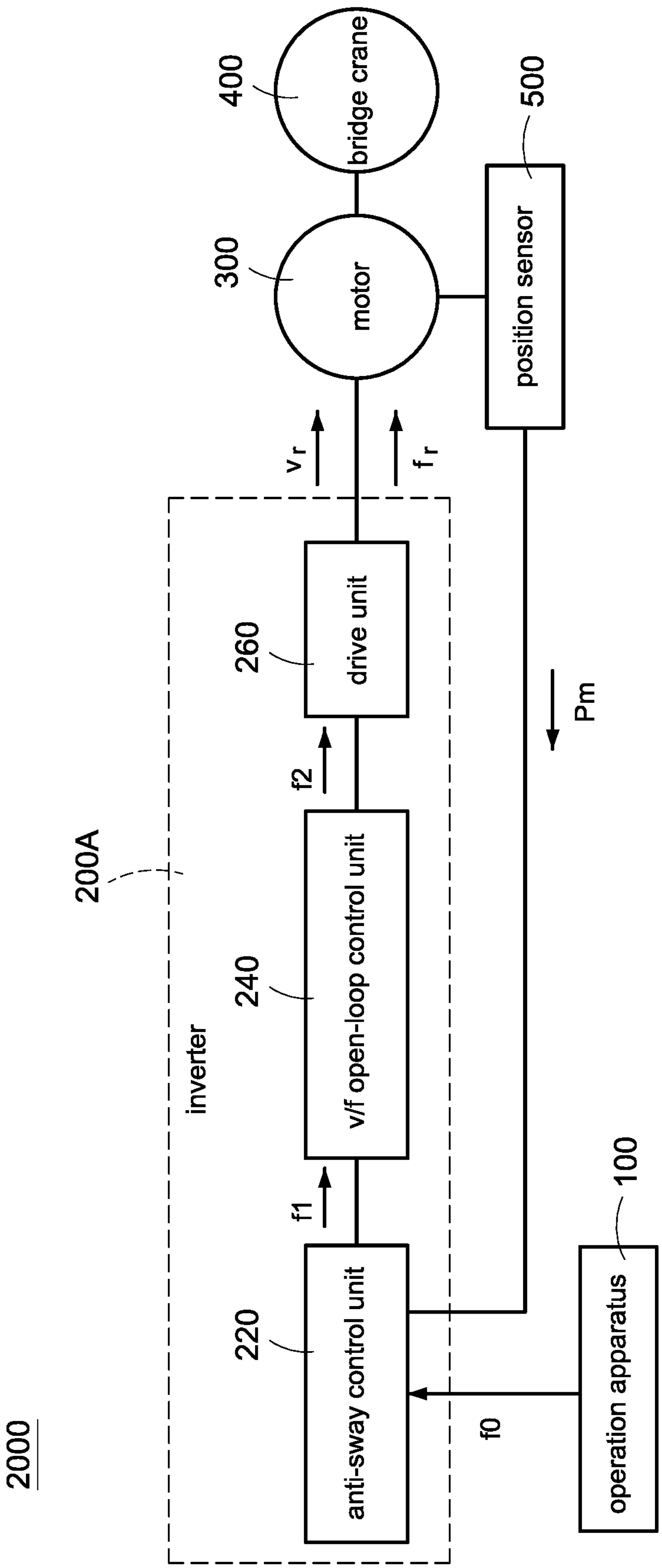

2000
200A
inverter
220
anti-sway control unit
f1
240
v/f open-loop control unit
f2
260
drive unit
v_r
f_r
300
motor
400
bridge crane
500
position sensor
Pm
100
operation apparatus
f0

FIG.8

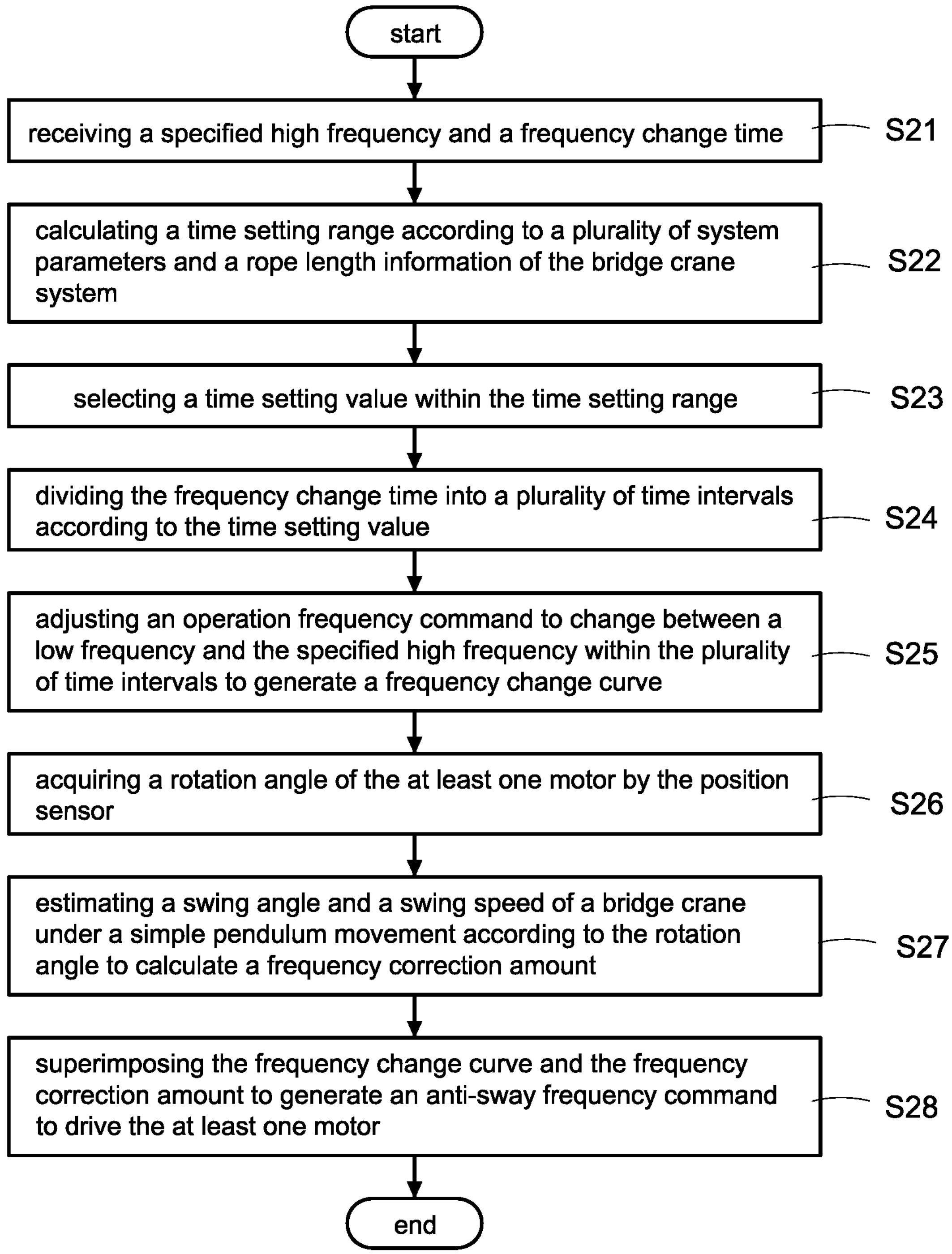

start
receiving a specified high frequency and a frequency change time
S21
calculating a time setting range according to a plurality of system parameters and a rope length information of the bridge crane system
S22
selecting a time setting value within the time setting range
S23
dividing the frequency change time into a plurality of time intervals according to the time setting value
S24
adjusting an operation frequency command to change between a low frequency and the specified high frequency within the plurality of time intervals to generate a frequency change curve
S25
acquiring a rotation angle of the at least one motor by the position sensor
S26
estimating a swing angle and a swing speed of a bridge crane under a simple pendulum movement according to the rotation angle to calculate a frequency correction amount
S27
superimposing the frequency change curve and the frequency correction amount to generate an anti-sway frequency command to drive the at least one motor
S28
end

FIG.9

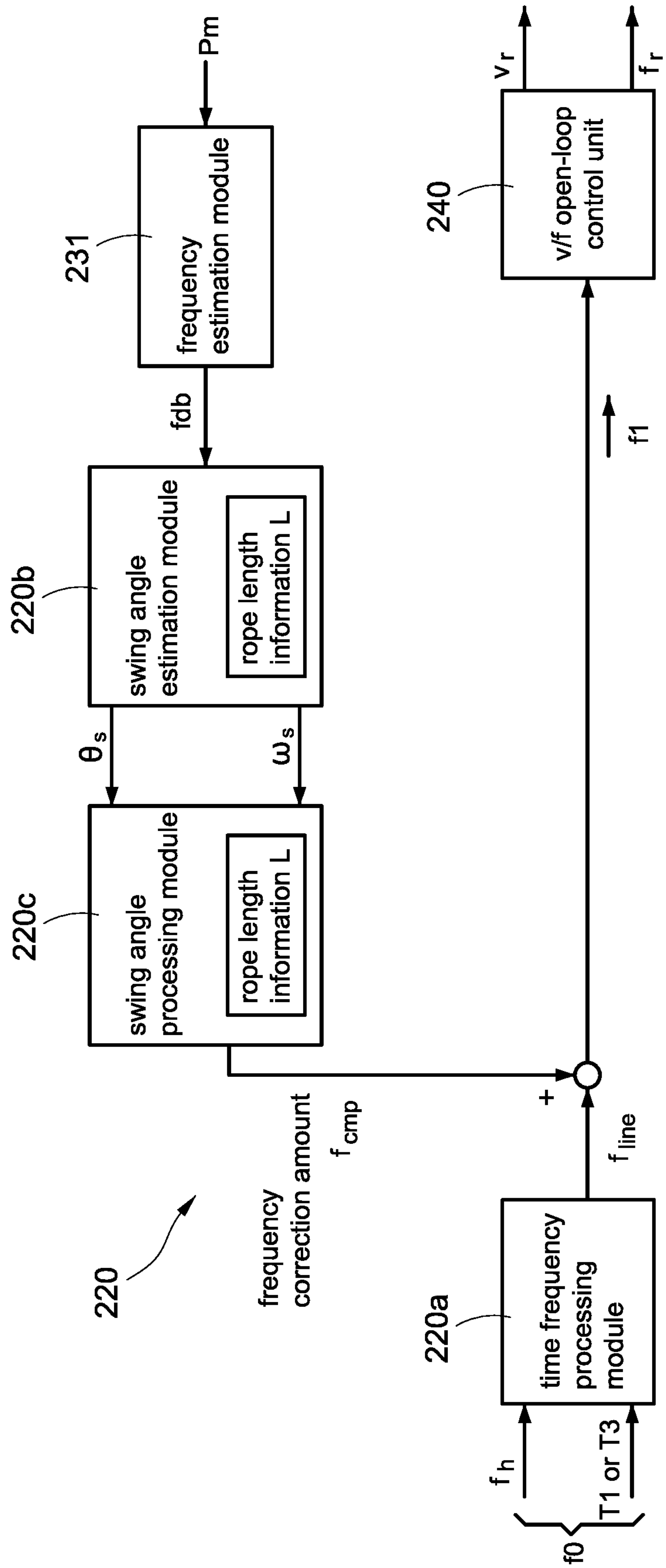

Pm
231
frequency estimation module
fdb
220b
swing angle estimation module
rope length information L
θ_s
ω_s
220c
swing angle processing module
rope length information L
220
frequency correction amount f_{cmp}
+
f_{line}
220a
time frequency processing module
f_h
T1 or T3
f0
f1
240
v/f open-loop control unit
v_r
f_r

FIG.10

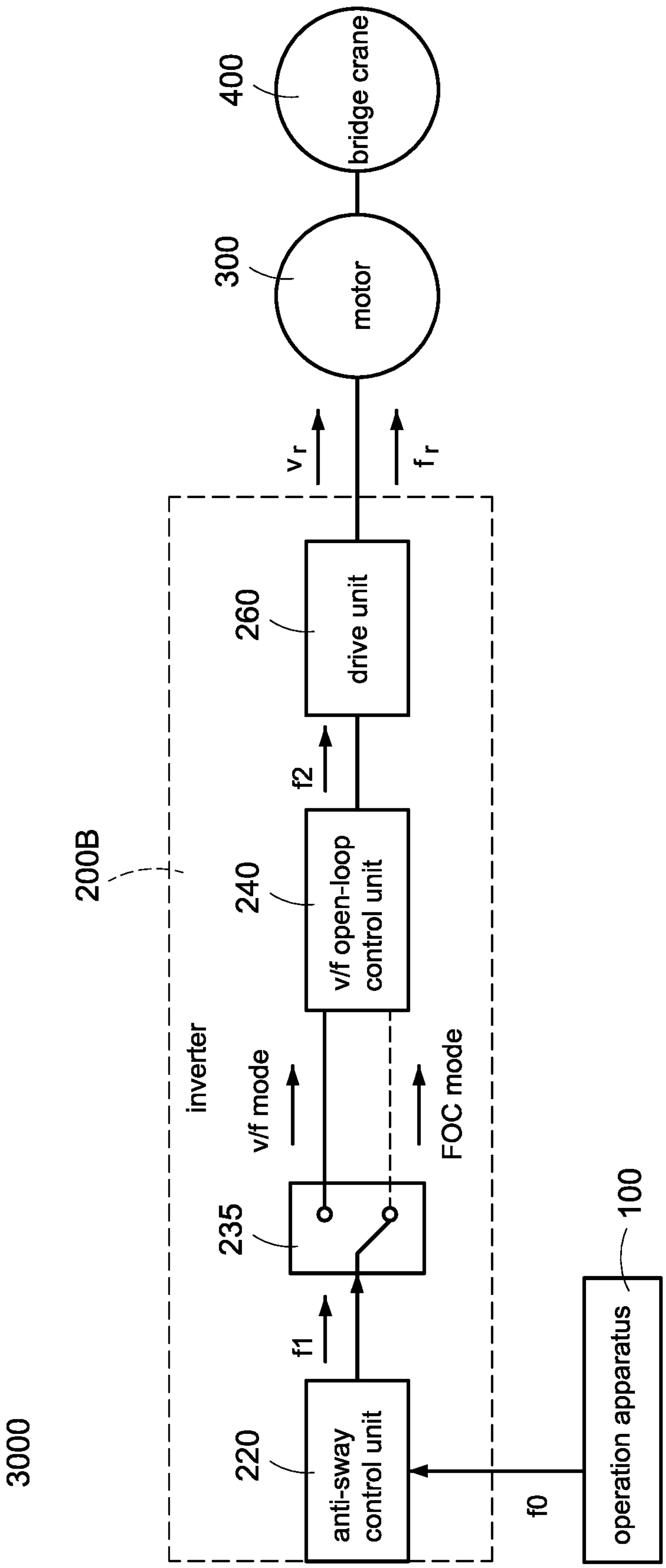

3000
200B
inverter
220
anti-sway control unit
f1
235
v/f mode
FOC mode
240
v/f open-loop control unit
f2
260
drive unit
v_r
f_r
300
motor
400
bridge crane
100
operation apparatus
f0

FIG.11

FULL-TIME ANTI-SWAY CONTROL METHOD OF BRIDGE CRANE SYSTEM BASED ON INVERTER STRUCTURE

CROSS-REFERENCE TO RELATED APPLICATION

This patent application claims the benefit of U.S. Provisional Patent Application No. 63/138,640, filed Jan. 18, 2021, which is incorporated by reference herein.

BACKGROUND

Technical Field

The present disclosure relates to a full-time anti-sway control method of a bridge crane system, and more particularly to a full-time anti-sway control method of a bridge crane system based on an inverter structure.

Description of Related Art

Bridge (overhead) cranes have been widely used in industrial assembly and transportation applications. A typical bridge crane structure includes a crane bridge, a crane trolley, and a hoist that moves up and down in a Z direction so that the hanging objects move to the designated position by the operation of the crane bridge and the crane trolley. However, during the operation process, the hanging object will inevitably sway due to the speed change of the crane bridge and/or the crane trolley, which affects work efficiency and increases work safety problems.

The anti-sway function for crane is suitable for indoor bridge crane facilities, which is used in the inverter structure of the crane bridge (in an X direction) and the crane trolley (in a Y direction). When the hoist suspends heavy objects and moves in the X or Y direction, the anti-sway function is activated/enabled to eliminate unnecessary swaying during the moving process, reduce the occurrence of hazards, increase production capacity, and achieve better bridge crane control benefits. Under the same number of operations, the operation time is a Gaussian distribution.

Many references have proposed related anti-sway technologies. Based on cost considerations, most of the anti-sway technologies use a swing angle estimator to replace an image identifier or (swing) angle sensor. Since the anti-sway controller adopts the design of state feedback, the design of the estimator and the state controller requires a large number of system parameters to be set. In practical applications, therefore, the system parameters are difficult to be measured and difficult to be acquired so as to increase the trouble of use.

Since it is necessary to estimate the speed of the crane bridge and the crane trolley as well as to set system parameters in terms of angle estimation, the use of motor position sensor is necessary. However, for low-cost system configurations, the motor may not be equipped with an encoder or Hall sensor, and even additional installation of the encoder or Hall sensor will increase the cost of mechanism design and hardware configuration, and also increase the difficulty of implementation.

In order to solve the above technical difficulties, the present disclosure proposes the full-time anti-sway control method of the bridge crane system based on the inverter structure, which is simple, easy to implement, without requiring a motor position sensor, and having low-cost hardware configuration.

SUMMARY

An object of the present disclosure is to provide a full-time anti-sway control method of a bridge crane system based on an inverter structure to solve the problems of existing technology.

In order to achieve the object of the present disclosure, the bridge crane system includes an inverter for performing the control method and at least one motor controlled by the control method. The control method includes steps of: receiving a specified high frequency and a frequency change time, calculating a time setting range according to a plurality of system parameters and a rope length information of the bridge crane system, selecting a time setting value within the time setting range, dividing the frequency change time into a plurality of time intervals according to the time setting value, adjusting an operation frequency command to change between a low frequency and the specified high frequency within the plurality of time intervals to generate a frequency change curve, calculating a frequency correction amount according to the frequency change curve and the rope length information, and superimposing the frequency change curve and the frequency correction amount to generate an anti-sway frequency command to drive the at least one motor.

In order to achieve the object of the present disclosure, the bridge crane system includes an inverter for performing the control method and at least one motor controlled by the control method. The control method includes steps of: receiving a specified high frequency and a frequency change time, calculating a time setting range according to a plurality of system parameters and a rope length information of the bridge crane system, selecting a time setting value within the time setting range, dividing the frequency change time into a plurality of time intervals according to the time setting value, adjusting an operation frequency command to change between a low frequency and the specified high frequency within the plurality of time intervals to generate a frequency change curve, acquiring a rotation angle of the at least one motor by the position sensor, estimating a swing angle and a swing speed of a bridge crane under a simple pendulum movement according to the rotation angle to calculate a frequency correction amount, and superimposing the frequency change curve and the frequency correction amount to generate an anti-sway frequency command to drive the at least one motor.

It is to be understood that both the foregoing general description and the following detailed description are exemplary, and are intended to provide further explanation of the present disclosure as claimed. Other advantages and features of the present disclosure will be apparent from the following description, drawings and claims.

BRIEF DESCRIPTION OF DRAWINGS

The present disclosure can be more fully understood by reading the following detailed description of the embodiment, with reference made to the accompanying drawing as follows:

FIG. **1** is a structure diagram of a full-time anti-sway control of a bridge crane system according to a first embodiment of the present disclosure.

FIG. **2** is a flowchart of a full-time anti-sway control method of the bridge crane system based on an inverter according to a first embodiment of the present disclosure.

FIG. **3** is a structure diagram of an anti-sway controller in FIG. **1**.

FIG. **4**A is a schematic block diagram of a swing angle estimation module of the full-time anti-sway control of the bridge crane system according to the first embodiment of the present disclosure.

FIG. **4**B is a schematic block diagram of a swing angle processing module of the full-time anti-sway control of the bridge crane system according to the first embodiment of the present disclosure.

FIG. **5**A is a schematic curve of an optimal frequency change of the full-time anti-sway control of the bridge crane system according to the first embodiment of the present disclosure.

FIG. **5**B is a schematic curve of an optimal anti-sway frequency command of the full-time anti-sway control of the bridge crane system according to the first embodiment of the present disclosure.

FIG. **6** is a schematic curve of the response of different damping coefficients to a swing angle and a motor speed according to the present disclosure.

FIG. **7** is a schematic curve of the response of different bandwidth ratios to the swing angle and the motor speed according to the present disclosure.

FIG. **8** is a structure diagram of the full-time anti-sway control of the bridge crane system according to a second embodiment of the present disclosure.

FIG. **9** is a flowchart of the full-time anti-sway control method of the bridge crane system based on the inverter according to a second embodiment of the present disclosure.

FIG. **10** is a structure diagram of the anti-sway controller in FIG. **8**.

FIG. **11** is a structure diagram of the bridge crane system having a drive mode switch according to the present disclosure.

FIG. **12** is a structure diagram of an anti-sway frequency controller having the drive mode switch according to the present disclosure.

DETAILED DESCRIPTION

Reference will now be made to the drawing figures to describe the present disclosure in detail. It will be understood that the drawing figures and exemplified embodiments of present disclosure are not limited to the details thereof. The present disclosure provides a full-time anti-sway control method of a bridge crane system, and the function of the method is based on an inverter, and the method has the following characteristics and functions.

Regardless of whether there is a motor position sensor or not, the anti-sway function can be completed without a swing angle sensor, and the construction cost is low. That is, no motor position sensor and swing angle sensor/image recognizer are needed, and the construction cost is low.

Only the rope length information is required, and the dependence on the parameters of the crane and motor system is low, and it is easy to implement. That is, there is no need to highly rely on system parameters such as the weight of the crane bridge and the crane trolley, the weight of the suspended objects, the wheel diameter, and the reduction ratio, and it is easy to implement.

In the crane bridge structure, if the inverter is used to drive two motors, simple v/f (voltage/frequency) control can also be used to achieve anti-sway function. That is, the v/f motor control method is suitable for one-to-many structure (multiple motors driven by one inverter), and it has high versatility.

The anti-sway frequency generator is generated all-time, regardless of the general travel (normal acceleration and deceleration in one direction), repeated inching (repeated acceleration and deceleration in one direction), repeated forward and reverse rotation (repeated forward movement and reverse movement) can achieve the anti-sway effect before the crane stops. That is, the full-time anti-sway control is suitable for all working conditions of the crane operation, and it has high versatility.

The control parameters can be automatically adjusted without repeated tests according to the intensity setting by the user. That is, it can perform full-time anti-sway and high control freedom.

Please refer to FIG. **1**. The bridge crane system **1000** includes an operation apparatus **100**, an inverter **200**, at least one motor **300**, and a bridge crane **400**. In one embodiment, the bridge crane **400** includes a crane bridge and a crane trolley. The bridge crane system **1000** uses a plurality of motors to drive different cranes of the bridge crane **400**. In other embodiments, the bridge crane system **1000** uses only one motor to drive the bridge crane **400** having a single crane. Therefore, the number of the motors is not limited in the present disclosure. In addition, the bridge crane **400** is usually controlled to perform a simple pendulum movement.

The user may use the operation apparatus **100**, such as a remote controller, a calculator, a computer, or so on to provide an operation command f**0** to the inverter **200**. The operation command f**0** includes information such as movement command, movement direction, given frequency, and acceleration/deceleration time, but the present disclosure is not limited thereto. As shown in FIG. **1**, the inverter **200** includes an anti-sway control unit **220**, a voltage/frequency (v/f) open-loop control unit **240**, and a drive unit **260**. The anti-sway control unit **220** outputs an anti-sway frequency command f**1** to the v/f open-loop control unit **240** according to the operation command f**0**. The v/f open-loop control unit **240** performs a voltage/frequency open-loop control (referred to as a v/f open-loop control) to the drive unit **260** according to the anti-sway frequency command f**1** so that the drive unit **260** generates a drive voltage signal v_r and a drive frequency signal f_r to drive (operate) the at least one motor **300**. In particular, the v/f open-loop control belongs to the technology well known to those of ordinary skill in the art, and therefore the detail description thereof is omitted here for conciseness. In general, the drive unit **260** may be a drive circuit with a plurality of switches, power converters, etc., but the present disclosure is not limited thereto.

Therefore, the focus of the present disclosure is how to generate the anti-sway frequency command f**1** to optimize the v/f open-loop control to reduce the sway phenomenon of the bridge crane. Please refer to FIG. **1**, FIG. **2**, FIG. **3**, FIG. **4**A, FIG. **4**B, FIG. **5**A, and FIG. **5**B to explain the control method of the first embodiment of the present disclosure.

In the step (S**11**) shown in FIG. **2**, a time frequency processing module **220**_a_ (shown in FIG. **3**) of the anti-sway control unit **220** receives the operation command f**0** provided from the operation apparatus **100**. In particular, the operation command f**0** includes, for example, but not limited to, a specified high frequency fh and a time setting value (such as an acceleration time T**1** or a deceleration time T**3** shown in FIG. **5**A), and a frequency change time T**0** is preset in the inverter **200**. In other embodiments, the operation command f**0** includes, for example, but not limited to, the specified high frequency fh, the time setting value (T**1** or T**3**), and/or the frequency change time T**0**, but the present disclosure is not limited thereto.

In the step (S**12**) shown in FIG. **2**, the time frequency processing module **220**_a_ calculates a time setting range according to a plurality of system parameters and a rope length information L of the bridge crane system **1000**. In one embodiment, the plurality of the system parameters includes a system inertia of the bridge crane system **1000**, and a rated speed and a rated torque of the motor **300**. In particular, the plurality of system parameters and the rope length information L of the bridge crane system **1000** are program setting values preset in the inverter **200**. In one embodiment, the present disclosure provides users to set the acceleration and deceleration time of the bridge crane system **1000** by themselves so that the acceleration and deceleration time of the bridge crane system **1000** can be flexibly designed under the conditions of inverter overcurrent limit and anti-sway control allowable time. However, the acceleration and deceleration time of the bridge crane system **1000** needs to be designed within a reasonable time setting range. The following continues to describe how to acquire the time setting range in the step (S**12**).

The time frequency processing module **220***a* calculates a lower limit value of the time setting range according to the system parameters of the bridge crane system **1000**. Please refer to the following equation (1).

$$t_{acc/dec} \geq \frac{J_{sys} \times \omega_{rate}}{2T_{rate}} \quad \text{equation (1)}$$

In the equation (1), $t_{acc/dec}$ is the time setting range; ω_{rate} is the rated speed; T_{rate} is the rated torque; J_{sys} is the system inertia. Afterward, the time frequency processing module **220***a* calculates the natural swing period of a simple pendulum according to the rope length information L of the bridge crane **400**. Please refer to the following equation (2).

$$T_{swing} = 2\pi\sqrt{\frac{L}{g}} \quad \text{equation (2)}$$

In the equation (2), T_{swing} is the natural swing period; g is the acceleration of gravity; L is the rope length information. Afterward, the time frequency processing module **220***a* calculates an upper limit value of the time setting range according to the natural swing period. Please refer to the following equation (3).

$$t_{acc/dec} \leq \frac{0.9T_{swing}}{2} \quad \text{equation (3)}$$

In the equation (3), $t_{acc/dec}$ is the time setting range; T_{swing} is the natural swing period. Therefore, the time setting range can be reasonably inferred by the combination of equations (1), (2), and (3).

In the step (S**13**) shown in FIG. **2**, the user may select the appropriate time setting values within the time setting range through the time frequency processing module **220***a*, and the selected time setting values are used as the acceleration and deceleration time of the bridge crane system **1000**. Please refer to FIG. **5**A, the selected time setting values are used as the acceleration time T**1** and the deceleration time T**3**. In one preferred embodiment, the acceleration time T**1** and the deceleration time T**3** are the same, but the present disclosure is not limited thereto.

In the step (S**14**) shown in FIG. **2**, the time frequency processing module **220***a* divides the frequency change time T**0** into multiple time intervals (as shown in FIG. **5**A) according to the time setting values selected in the step (S**12**). The multiple time intervals include the acceleration time T**1**, the maintain time T**2**, and the deceleration time T**3**. In one embodiment, as shown in FIG. **5**A, the selected time setting values may be used as the acceleration time T**1** and the deceleration time T**3**, and the acceleration time T**1** and the deceleration time T**3** are the same. Therefore, the maintain time T**2** can be inferred based on the frequency change time T**0**, the acceleration time T**1**, and the deceleration time T**3**. In other words, multiple maintain times T**2** within the multiple time intervals can be acquired according to the frequency change time T**0** and the time setting values. The multiple maintain times T**2** are between the acceleration time T**1** and the deceleration time T**3**. In particular, in the present disclosure, the user selects the time setting values within the time setting range to set the acceleration time T**1** and the deceleration time T**3**. However, the user cannot operate the time of starting or stopping the bridge crane system **1000**.

In the step (S**15**) shown in FIG. **2**, the time frequency processing module **220***a* adjusts an operation frequency command to change between a low frequency (such as 0 Hz) and the specified high frequency f_h within multiple time intervals T**1**-T**3** so as to generate a frequency change curve f_{line} (as shown in FIG. **5**A). In particular, the operation frequency command is a signal generated in the time frequency processing module **220***a* in advance, or a signal preset in the time frequency processing module **220***a*.

In one embodiment, as shown in FIG. **5**A, in the acceleration time T**1**, the frequency change curve f_{line} linearly increases from the low frequency to the specified high frequency f_h. In the maintain time T**2**, the frequency change curve f_{line} maintains at the specified high frequency f_h. In fact, frequency change curve f_{line} oscillates within an error range of the specified high frequency f_h. In the deceleration time T**3**, the frequency change curve f_{line} linearly decreases from the specified high frequency f_h to the low frequency, but the present disclosure is not limited thereto. In other words, the frequency change curve f_{line} linearly increases from the low frequency (such as **0** Hz) to the specified high frequency f_h, maintains at the specified high frequency f_h, and linearly decreases from the specified high frequency f_h to the low frequency.

As shown in FIG. **3**, the anti-sway control unit **220** includes a swing angle estimation module **220***b* and a swing angle processing module **220***c*. Please refer to FIG. **2** and FIG. **3**, in the step (S**16**), the swing angle estimation module **220***b* and the swing angle processing module **220***c* calculate a frequency correction amount f_{cmp} according to the frequency change curve f_{line} and the preset rope length information L. The calculation method of the frequency correction amount f_{cmp} will be detailed below.

Please refer to FIG. **2** (S**16**), FIG. **3**, and FIG. **4**A, the swing angle estimation module **220***b* firstly receives the frequency change curve f_{line} outputted from the time frequency processing module **220***a*. The swing angle estimation module **220***b* calculates a frequency change amount of the operation frequency command within the frequency change time T**0** according to the frequency change curve f_{line}. Please refer to FIG. **5**A, the frequency on the vertical axis represents the speed, and the acceleration (that is, the frequency change amount) may be acquired by differentiating the speed once (i.e., the frequency change amount). However, the calculation method of the frequency change amount of the present disclosure is not limited to the above-mentioned method.

Afterward, the swing angle estimation module **220***b* calculates a swing angle θ_s of the bridge crane **400** under the simple pendulum movement according to the rope length information L and the frequency change amount. Please refer to the following equation (4).

$$\theta_s = \frac{1}{s^2 + G/L} \times \frac{\Delta f}{L} \quad \text{equation (4)}$$

In the equation (4), θ_s is the swing angle; s is a Laplace operator; G is a gravitational acceleration constant; L is the rope length information; Δf is the frequency change amount. Afterward, the swing angle estimation module **220***b* calculates a swing speed ω_s of the bridge crane **400** under the simple pendulum movement according to the swing angle θ_s. In some embodiments, the swing angle estimation module **220***b* differentiates the swing angle θ_s to acquire the swing speed ω_s of the bridge crane **400** under the simple pendulum movement. The swing angle estimation module **220***b* provides the swing angle θ_s and the swing speed ω_s to the swing angle processing module **220***c*.

Please refer to FIG. **2** (S**15**), FIG. **3**, and FIG. **4**B, the swing angle processing module **220***c* includes a first control parameter C and a second control parameter γ. The swing angle processing module **220***c* firstly calculates a control variable X**1** according to the swing angle θ_s, the swing speed ω_s, and the first control parameter C. Please refer to the following equation (5).

$$X1 = C\theta_s + \omega_s \quad \text{equation(5)}$$

Afterward, the swing angle processing module **220***c* multiplies the control variable X**1** with the second control parameter γ to calculate a rotation speed correction amount ω_{cmp}. Please refer to the following equation (6).

$$\omega_{cmp} = \gamma \times X1 \quad \text{equation(6)}$$

After the swing angle processing module **220***c* calculates the rotation speed correction amount ω_{cmp}, the frequency correction amount f_{cmp} can be calculated according to the following equation (7).

$$f_{cmp} = \frac{\omega_{cmp} \times P}{120} \quad \text{equation (7)}$$

The following will continue to introduce an embodiment of the present disclosure for designing the first control parameter C and the second control parameter γ. In general, the bridge crane system **1000** may be simplified into a second-order control system, as the following equation (8).

$$\frac{\theta_s}{f^*} = \frac{\frac{s}{L(\gamma+1)}}{s^2 + \frac{L\gamma C}{L(\gamma+1)}s + \frac{g}{L(\gamma+1)}} = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_2^2} \quad \text{equation(8)}$$

In the equation (8), L is the rope length information; θ_s is the swing angle; f* is a frequency command; ζ is a damping coefficient; ω_n is a bandwidth. In particular, the frequency command (f*) is the equation (8) is the frequency change curve f_{line} in FIG. **5**A. Afterward, the equation (8) is transformed into a standard second-order equation to derive the equation (9) and the equation (10) as follows.

$$2\zeta\omega_n = \frac{L\gamma C}{L(\gamma+1)} \quad \text{equation (9)}$$

$$\omega_n^2 = \frac{G}{L(\gamma+1)} \quad \text{equation (10)}$$

After adjusting the equation (9) and the equation (10), the first control parameter C and the second control parameter γ can be inferred, as shown in the following equation (11) and equation (12).

$$\gamma = \frac{G}{L\omega_n^2} - 1 > 0 \quad \text{equation (11)}$$

$$C = \frac{2\zeta\omega_n G}{G - \omega_n^2 L} > 0 \quad \text{equation (12)}$$

According to the equation (11) and the equation (12), the first control parameter C and the second control parameter γ can be acquired by designing different damping coefficients ζ and bandwidths ω_n. In one preferred embodiment, a range of the damping coefficient ζ is between 0.1 and 1 ($\zeta \in (0.1, 1)$), and the bandwidth ω_n is shown in the following equation (13).

$$\omega_n = n \times \omega_{swing} = n \times \sqrt{G/L} \quad \text{equation(13)}$$

In the equation (13), ω_{swing} is a swing frequency of the bridge crane **400**; n is a bandwidth ratio. In particular, the swing frequency ω_{swing} of the bridge crane **400** is derived by the natural swing period T_{swing} (such as the equation (2)) of the bridge crane **400**.

In the present disclosure, the damping coefficient ζ and the bandwidth ratio n are designed by users to adjust to meet the requirements of controlling the system. In particular, by designing different damping coefficients ζ, the anti-sway system rigidity can be adjusted, and by designing different bandwidth ratios n, the response speed (strength) can be adjusted.

As shown in FIG. **6**, which shows a schematic curve of the response of different damping coefficients to a swing angle and a motor speed according to the present disclosure. In FIG. **8**, when the damping coefficient ζ is smaller, the degree of suppression of the swing angle is smaller, and the maximum overshoot of the motor speed is larger. On the contrary, when the damping coefficient ζ is larger, the degree of suppression of the swing angle is larger, and the maximum overshoot of the motor speed is smaller. Therefore, when the motor decelerates, the larger the damping coefficient ζ, the less likely it is to reverse rotation. In general, the damping coefficient ζ may be set at a relatively intermediate value at the factory (ζ=0.707). However, the design of the damping coefficient ζ may be adjusted according to the user's operating habits and preferences. Since the effect of the damping coefficient ζ may be compared to the shock-absorbing effect of a vehicle, if the damping coefficient ζ is small, the amount of shaking will be more obvious.

As shown in FIG. **7**, which shows a schematic curve of the response of different bandwidth ratios to the swing angle and the motor speed according to the present disclosure. In FIG. **9**, when the bandwidth ratio n is larger, an anti-sway time (that is, the time from the start of deceleration to the speed reaching a steady state) is relatively small. On the contrary, when the bandwidth ratio n is smaller, the anti-sway time is relatively large. Under the principle of reasonable swing angle, the larger the bandwidth ratio n, the faster the speed of compensating the swing angle (zero), i.e., returning to the steady state faster), and the swing will be more severe.

By the combination of equations (2), (11), (12), and (13), the design method of the first control parameter C and the second control parameter γ has the following steps. According to the natural swing period T_{swing}, the response frequency (referred to as the bandwidth ω_n) is calculated. According to the response frequency (i.e., the bandwidth ω_n), the damping coefficient ζ, and the rope length information L, the first control parameter C is calculated. According to the response frequency (i.e., the bandwidth ω_n) and the rope length information L, the second control parameter γ is calculated. In particular, the first control parameter C and the second control parameter γ may be preset in the swing angle processing module **220***c* after calculation by the user, but the present disclosure is not limited thereto.

Please refer to FIG. **1**, FIG. **2** (S**17**), FIG. **3**, and FIG. **5**B, the inverter **200** superimposes the frequency change curve fine and the frequency correction amount f_{cmp} generated by the swing angle processing module **220***c* to generate an anti-sway frequency command **f1** to drive the at least one motor **300**. As shown in FIG. **5**B, since the frequency change curve fine and the frequency correction amount f_{cmp} are superimposed to generate the anti-sway frequency command **f1**, the anti-sway frequency command **f1** nonlinearly increases from a low frequency (for example, but not limited to 0 Hz) to a specified high frequency f_h. After the anti-sway frequency command **f1** maintains at the specified high frequency f_h for a period of time, the anti-sway frequency command **f1** nonlinearly decreases from the specified high frequency f_h to the low frequency.

Please refer to FIG. **8**, the bridge crane system **2000** includes an operation apparatus **100**, an inverter **200**A, at least one motor **300**, a bridge crane **400**, and a position sensor **500**. The inverter **200**A includes an anti-sway control unit **230**, a voltage/frequency (v/f) open-loop control unit **240**, and a drive unit **260**. Please refer to FIG. **10**, the anti-sway control unit **230** includes a time frequency processing module **220***a*, a frequency estimation module **231**, a swing angle estimation module **220***b*, and a swing angle processing module **220***c*. In the second embodiment, the position sensor **500** is used to detect/sense the motor **300**, and outputs a motor position signal Pm to the anti-sway control unit **230**. The anti-sway control unit **230** generates an anti-sway frequency command **f1** to the v/f open-loop control unit **240** according to the motor position signal Pm. Please refer to FIG. **8**, FIG. **9**, and FIG. **10** to explain the control method of the second embodiment of the present disclosure.

In the second embodiment, the time frequency processing module **220***a* performs the steps (S**21**) to (S**25**) to generate the frequency change curve fine (as shown in FIG. **5**A). In particular, the operation method of steps (S**21**) to (S**25**) in FIG. **9** is the same as that of steps (S**11**) to (S**15**) in FIG. **2**, and the detail description is omitted here for conciseness.

Please refer to FIG. **9** (S**26**) and FIG. **10**, the anti-sway control unit **230** acquires a rotation angle of the motor **300** according to the motor position signal Pm outputted from the position sensor **500**.

Please refer to FIG. **9** (S**27**) and FIG. **10**, the anti-sway control unit **230** estimates the swing angle θ_s and the swing speed ω_s of the bridge crane **400** under the simple pendulum movement according to the rotation angle of the motor **300** so as to calculate the frequency correction amount f_{cmp}. In this embodiment, the frequency estimation module **231** differentiates the rotation angle of the motor **300** to acquire the rotation speed of the motor **300**, and uses the rotation speed of the motor **300** as an electrical frequency command fdb. The frequency estimation module **231** outputs the electrical frequency command fdb to the swing angle estimation module **220***b*.

Afterward, the swing angle estimation module **220***b* differentiates the electrical frequency command fdb once to acquire the frequency change amount (i.e., the acceleration) so as to estimate the swing angle θ_s and the swing speed ω_s. The method of estimating the swing angle θ_s has been described in the previous paragraph, please refer to equation (4). The swing angle estimation module **220***b* differentiates the swing angle θ_s to acquire the swing speed ω_s. Afterward, the swing angle estimation module **220***b* outputs the swing angle θ_s and the swing speed ω_s to the swing angle processing module **220***c*. The swing angle processing module **220***c* calculates the frequency correction amount f_{cmp} according to the above-mentioned equations (5) to (13).

Please refer to FIG. **9** (S**28**) and FIG. **10**, the inverter **200**A superimposes the frequency change curve fine and the frequency correction amount f_{cmp} generated by the swing angle processing module **220***c* to generate an anti-sway frequency command **f1** to drive the at least one motor **300**. In particular, the waveform of the anti-sway frequency command **f1** is shown in FIG. **5**B.

The anti-sway control structure of the present disclosure may also use a drive mode switch **235** to select the power unit applied to the motor with/without the position sensor to configure to the bridge crane structure, and to switch the drive mode according to the actual hardware configuration to provide the flexibility in use, as shown in FIG. **11**.

In summary, the anti-sway control structure proposed in the present disclosure may be used for the bridge crane system having the motor without position sensor, such as an incremental encoder, an absolute encoder, or a Hall sensor, and swing angle sensor, such as an angle sensor, a gyroscope, an inclinometer, and an image recognizer. In addition, the anti-sway control structure proposed in the present invention does not require too many other bridge crane system parameters (for example, the equivalent radius of the rotation to line, and the number of motor rotors). The present disclosure only needs low-cost power unit configuration and v/f drive control mode to linearly move the anti-sway control structure, and can perform full-time anti-sway during operation, even if the inching movement command is executed, it also has the effect of the anti-sway control. Whether using the v/f control or the vector control (FOC) requiring rotor information, the full-time anti-sway performance can be achieved through the control method of the present disclosure. The present disclosure uses the motor input frequency as the input source of the swing angle estimation module **220***b* (or referred to as a swing angle estimator) under the v/f control so that the anti-sway control unit has the characteristics of stability and no additional filter design is required.

The swing angle estimation module **220***b* of the present disclosure can estimate the swing angle of the hanging object or the hook without relying on the weight of the long travel, the trolley, and the hanging object, and without the gear ratio of the reduction box and the wheel diameters of the long travel and the trolley.

Although the present disclosure has been described with reference to the preferred embodiment thereof, it will be understood that the present disclosure is not limited to the details thereof. Various substitutions and modifications have been suggested in the foregoing description, and others will occur to those of ordinary skill in the art. Therefore, all such substitutions and modifications are intended to be embraced within the scope of the present disclosure as defined in the appended claims.

What is claimed is:

1. A full-time anti-sway control method of a bridge crane system based on an inverter structure, the bridge crane system comprising an inverter for performing the control method and at least one motor controlled by the control method, the control method comprising steps of:

receiving a specified high frequency command and a frequency change time, wherein the frequency change time is an acceleration and deceleration time,

calculating a time setting range according to a plurality of system parameters and a rope length information of the bridge crane system, wherein the time setting range is the acceleration and deceleration time, wherein

$$t_{acc/dec} \geq \frac{J_{sys} \times \omega_{rate}}{2T_{rate}} \text{ and } t_{acc/dec} \leq \frac{0.9T_{swing}}{2},$$

where $t_{acc/dec}$ is the time setting range, ω_{rate} is a rated speed, T_{rate} is a rated torque, J_{sys} is a system inertia, T_{swing} is a natural swing period,

selecting a time setting value within the time setting range,

dividing the frequency change time into a plurality of time intervals according to the time setting value,

adjusting an operation frequency command to change between a low frequency command and the specified high frequency command within the plurality of time intervals to generate a frequency change curve,

estimating a swing angle and a swing speed by a swing angle estimation module in a sensorless manner according to the frequency change curve, a gravitational acceleration constant, and the rope length information,

designing a swing angle processing module according to the gravitational acceleration constant and the rope length information,

generating a frequency correction amount by the swing angle processing module according to the swing angle and the swing speed, and

superimposing the frequency change curve and the frequency correction amount to generate an anti-sway frequency command to drive the at least one motor.

2. The full-time anti-sway control method as claimed in claim **1**, further comprising steps of:

calculating a lower limit value of the time setting range according to the system parameters of the bridge crane system,

calculating a natural swing period of a bridge crane under a simple pendulum movement according to the rope length information of the bridge crane system, and

calculating an upper limit value of the time setting range according to the natural swing period.

3. The full-time anti-sway control method as claimed in claim **1**, further comprising steps of:

using the time setting value as an acceleration time and a deceleration time in the plurality of time intervals, and

acquiring a maintain time of the plurality of time intervals according to the frequency change time and the time setting value, wherein the maintain time is between the acceleration time and the deceleration time.

4. The full-time anti-sway control method as claimed in claim **3**, wherein the frequency change curve linearly increases from the low frequency command to the specified high frequency command, and after the frequency change curve maintains at the maintain time with the specified high frequency command, the frequency change curve linearly decreases from the specified high frequency command to the low frequency command.

5. The full-time anti-sway control method as claimed in claim **2**, further comprising:

calculating a frequency change amount of the operation frequency command within the frequency change time according to the frequency change curve,

calculating the swing angle of the bridge crane under the simple pendulum movement according to the rope length information and the frequency change amount, and

calculating the swing speed of the bridge crane under the simple pendulum movement according to the swing angle.

6. The full-time anti-sway control method as claimed in claim **5**, further comprising:

calculating a response frequency according to the natural swing period and a bandwidth ratio,

calculating a first control parameter according to the response frequency, a damping coefficient, and the rope length information,

calculating a second control parameter according to the response frequency and the rope length information,

calculating a control variable according to the swing angle, the swing speed, and the first control parameter,

multiplying the control variable with the second control parameter to acquire a rotation speed correction amount, and

acquiring the frequency correction amount according to the rotation speed correction amount.

7. The full-time anti-sway control method as claimed in claim **3**, wherein the anti-sway frequency command nonlinearly increases from the low frequency command to the specified high frequency command, and after the anti-sway frequency command maintains at the specified high frequency command for a period of time, the anti-sway frequency command nonlinearly decreases from the specified high frequency command to the low frequency command.

8. A full-time anti-sway control method of a bridge crane system based on an inverter structure, the bridge crane system comprising an inverter for performing the control method, a position sensor, and at least one motor controlled by the control method, the control method comprising steps of:

receiving a specified high frequency command and a frequency change time, wherein the frequency change time is an acceleration and deceleration time,

calculating a time setting range according to a plurality of system parameters and a rope length information of the bridge crane system, wherein the time setting range is the acceleration and deceleration time, wherein

$$t_{acc/dec} \geq \frac{J_{sys} \times \omega_{rate}}{2T_{rate}} \text{ and } t_{acc/dec} \leq \frac{0.9T_{swing}}{2},$$

where $t_{acc/dec}$ is the time setting range, ω_{rate} is a rated speed, T_{rate} is a rated torque, J_{sys} is a system inertia, T_{swing} is a natural swing period,

selecting a time setting value within the time setting range,

dividing the frequency change time into a plurality of time intervals according to the time setting value,

adjusting an operation frequency command to change between a low frequency command and the specified high frequency command within the plurality of time intervals to generate a frequency change curve,

acquiring a rotation angle of the at least one motor by the position sensor,

estimating a swing angle and a swing speed by a swing angle estimation module in a sensorless manner according to the frequency change curve, a gravitational acceleration constant, and the rope length information,

designing a swing angle processing module by the gravitational acceleration constant and the rope length information,

generating a frequency correction amount by the swing angle processing module according to the swing angle and the swing speed, and

superimposing the frequency change curve and the frequency correction amount to generate an anti-sway frequency command to drive the at least one motor.

9. The full-time anti-sway control method as claimed in claim **8**, further comprising steps of:

calculating a lower limit value of the time setting range according to the system parameters of the bridge crane system,

calculating a natural swing period of the bridge crane under the simple pendulum movement according to the rope length information of the bridge crane system, and

calculating an upper limit value of the time setting range according to the natural swing period.

10. The full-time anti-sway control method as claimed in claim **8**, further comprising steps of:

using the time setting value as an acceleration time and a deceleration time in the plurality of time intervals, and

acquiring a maintain time of the plurality of time intervals according to the frequency change time and the time setting value, wherein the maintain time is between the acceleration time and the deceleration time.

11. The full-time anti-sway control method as claimed in claim **10**, wherein the frequency change curve linearly increases from the low frequency command to the specified high frequency command, and after the frequency change curve maintains at the maintain time with the specified high frequency command, the frequency change curve linearly decreases from the specified high frequency command to the low frequency command.

12. The full-time anti-sway control method as claimed in claim **9**, further comprising:

calculating a frequency change amount of the operation frequency command within the frequency change time according to the frequency change curve,

calculating the swing angle of the bridge crane under the simple pendulum movement according to the rope length information and the frequency change amount, and

calculating the swing speed of the bridge crane under the simple pendulum movement according to the swing angle.

13. The full-time anti-sway control method as claimed in claim **12**, further comprising:

calculating a response frequency according to the natural swing period and a bandwidth ratio,

calculating a first control parameter according to the response frequency, a damping coefficient, and the rope length information,

calculating a second control parameter according to the response frequency and the rope length information,

calculating a control variable according to the swing angle, the swing speed, and the first control parameter,

multiplying the control variable with the second control parameter to acquire a rotation speed correction amount, and

acquiring the frequency correction amount according to the rotation speed correction amount.

14. The full-time anti-sway control method as claimed in claim **10**, wherein the anti-sway frequency command nonlinearly increases from the low frequency command to the specified high frequency command, and after the anti-sway frequency command maintains at the specified high frequency command for a period of time, the anti-sway frequency command nonlinearly decreases from the specified high frequency command to the low frequency command.

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